ADMINISTRATIVE NOTICE

NOT MEASUREMENT SENSITIVE

JOINT-UXS-002
NOTICE 1
20 January 2022

MODULAR OPEN SYSTEMS APPROACH (MOSA) PORTFOLIO – HARDWARE OPEN SYSTEMS TECHNOLOGIES (HOST), FUTURE AIRBORNE CAPABILITY ENVIRONMENT™ OR FACE™, SENSOR OPEN SYSTEMS ARCHITECTURE OR SOSA™, UNMANNED SYSTEMS (UXS) CONTROL SEGMENT (UCS), AND OPEN MISSION SYSTEMS (OMS)

MODULAR OPEN SYSTEMS APPROACH (MOSA) – STANDARDS ADOPTION

The U.S. Navy, U.S. Air Force, and U.S. Army have continued to push forward with the development and adoption of MOSA. The ground-based MOSA VICTORY Administrative Notice is released via ASSIST as ARMY-VICTORY-001. Mature Open Architecture (OA) standards, and alignment between standards, has allowed airborne systems to maximize the benefits of MOSA. This MOSA portfolio supports airborne systems, with many successful implementations of unmanned systems. The following five MOSA standards demonstrate applications from a low level viewpoint to higher level viewpoints: Hardware Open Systems Technologies (HOST), Future Airborne Capability Environment or FACE, Sensor Open Systems Architecture or SOSA, Unmanned Systems (UxS) Control Segment (UCS), and Open Mission Systems (OMS).

STANDARD SPECIFICATION

HARDWARE OPEN SYSTEMS TECHNOLOGIES (HOST)

HOST is a standards framework that defines multiple OAs that support a MOSA and standardizes the implementation of Commercial-Off-the-Shelf (COTS) components for embedded computing on open systems for U.S. defense platforms.

HOST enables the development of interoperable, modular, and upgradeable systems across military platforms and leverages open standards across product families to improve defense acquisition.

More information regarding the HOST standards framework can be found via https://host-oa.com/.

STANDARD SPECIFICATION

FUTURE AIRBORNE CAPABILITY ENVIRONMENT – THE OPEN GROUP FACE CONSORTIUM

The Open Group FACE Consortium is a Government and industry partnership, which also includes academia participation, to define an open avionics environment for all military airborne platform types. The FACE Consortium works to develop and consolidate the open standards, best practices, guidance documents, and business models to achieve the following objectives:

- Standardized approaches for using open standards for software development and integration
- Lower implementation costs of FACE systems
- Standards that support a robust architecture and enable quality software development
- The use of standard interfaces that will lead to the reuse of capabilities
- Portability of applications across multiple FACE systems and vendors

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:30Z
Check the source to verify that this is the current version before use.

- Procurement of FACE conformant products
- More capabilities reaching the warfighter faster
- Innovation and competition within Department of Defense (DoD) acquisition
- Robust, interoperable, portable, and secure computing platforms

More information regarding the FACE technical standard can be found via https://www.opengroup.org/face.

STANDARD SPECIFICATION

SENSOR OPEN SYSTEMS ARCHITECTURE

The Open Group SOSA Consortium enables Government and industry to collaboratively develop open standards and best practices to enable, enhance, and accelerate the deployment of affordable, capable, and interoperable sensor systems. The SOSA Consortium is creating open system reference architectures applicable to military and commercial sensor systems and a business model that balances stakeholder interests. The architectures employ modular design and use widely supported, consensus-based, nonproprietary standards for key interfaces that are expected to:

- Reduce development cycle time and cost
- Reduce systems integration cost and risk
- Increase commonality and reuse
- Reduce sustainment and modernization cost
- Support capability evolution and mitigate obsolescence
- Enable technology transition
- Facilitate interoperability
- Isolate the effects of change

More information regarding the SOSA aligned/upcoming technical standard can be found via https://www.opengroup.org/sosa.

STANDARD SPECIFICATION

UNMANNED SYSTEMS (UxS) CONTROL SEGMENT (UCS) – SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

The UCS architecture was started out of the Office of Secretary of Defense (OSD) Acquisition, Technology, and Logistics (AT&L), mandated across the Tri-Services for all Unmanned Aircraft Systems (UAS), and later adopted by the Navy for all UxS within its portfolio. The UCS standard was developed with DoD services and industry partners and transitioned to the SAE AS-4 UxS, Aerospace Standard (AS) working group for continued collaboration across all UxS stakeholders. The UCS architecture provides a commercially available and used Unified Modeling Language (UML) as a Service Oriented Architecture (SOA), which includes a data dictionary, a modeling framework, a specification, integration capabilities, and implementation for deployment of all UxS software across the Tri-Services. The UCS open standard architecture has the following goals:

- Enabling small business competition
- Increased capabilities while achieving rapid development and cost reduction across development, sustainment, and life-cycle management
- Some platform independence, which enables UCS architecture to be leveraged across different computing infrastructures and different communication protocols (e.g., Air Forces OMS, Joint Architecture for Unmanned Systems [JAUS], FACE, and Joint All-Domain Command and Control [JADC2])

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:30Z
Check the source to verify that this is the current version before use.

- Successful reuse of software and commonality across Tri-Service platforms

UCS technical standards, based on SAE AS6518A, can be found via https://www.sae.org/standards/content/as6518a/.

STANDARD SPECIFICATION

OPEN MISSION SYSTEMS (OMS)

The goal of the OMS is to develop industry and DoD consensus for a non-proprietary mission system architectural standard that enables affordable technical refresh and insertion, simplified mission systems integration, service reuse and interoperability, and competition across the life cycle. Industry and the Government have been working on solutions cooperatively to achieve these goals for several years with extensive testing and demonstrations.

The OMS standard can be found via https://www.vdl.afrl.af.mil/programs/oam/oms.php.

OTHER STANDARDS AND PATH FORWARD

The standards here are a breadth of today's leveraged standards. The Navy will continue on its path by leading the way in reviewing, developing, and participating in future standards as it pertains to benefiting rapid development, cost reduction, and interoperability across its portfolio with new capabilities for tomorrow's warfighter.

CONCLUDING MATERIAL

Custodians:
Army – OCSE
Navy – SH
Air Force – 10
OSD – SE

Preparing activity:
Navy – SH

Review activities:
Army – MI
Navy – AS, CG, CH, EC, MC, NM, OS, SA
Air Force – 01
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:30Z
Check the source to verify that this is the current version before use.